DATA ANALYSIS REPORT

CUSTOMER 360

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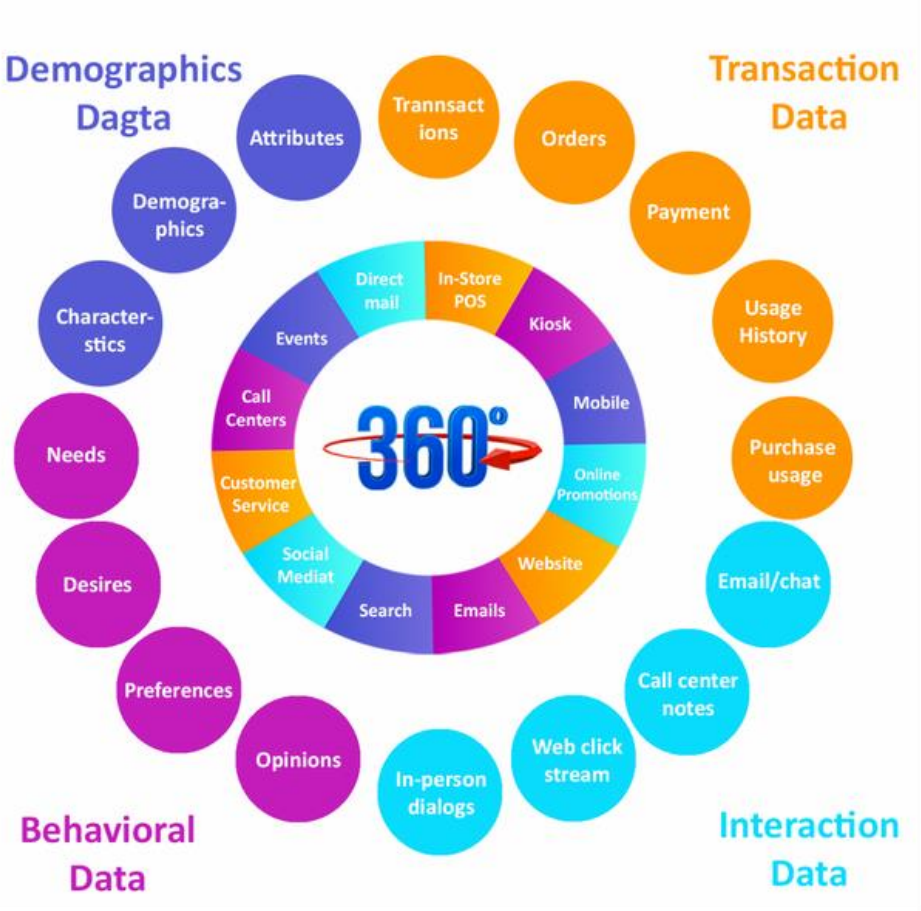
I. OVERVIEW OF CUSTOMER 360 & RFM ANALYSIS

1. Introducing Customer 360

Customer 360 provides a holistic view of each customer by integrating data from multiple sources—such as transactions, behavior, interaction and demographics — into one unified profile. It helps organizations understand customer journeys, preferences, and value over time.



2. Benefits of Customer 360

- **Unified Customer View**: Customer 360 brings all customer information together in one place. Everyone in the company can see the same data and understand the customer better. This helps avoid confusion and builds a clear picture of each customer.

- **Personalized Experience**: It helps businesses know what each customer likes and needs. With this knowledge, they can send the right messages, offers, and services. Customers feel more valued and happy with the brand.

- **Improved Customer Retention**: Customer 360 shows which customers may stop buying soon. Businesses can take quick action to keep them, such as giving special offers or support. This helps build stronger, longer relationships.

- **Smarter Decision-Making**: When all customer data is in one place, companies can make better and faster decisions. They can see what works and what doesn't. This helps improve marketing, sales, and customer service plans.

- **Enhanced Collaboration**: All teams—marketing, sales, and service—can use the same information. This makes it easier to work together and serve customers better. Everyone stays connected and focused on the same goals.

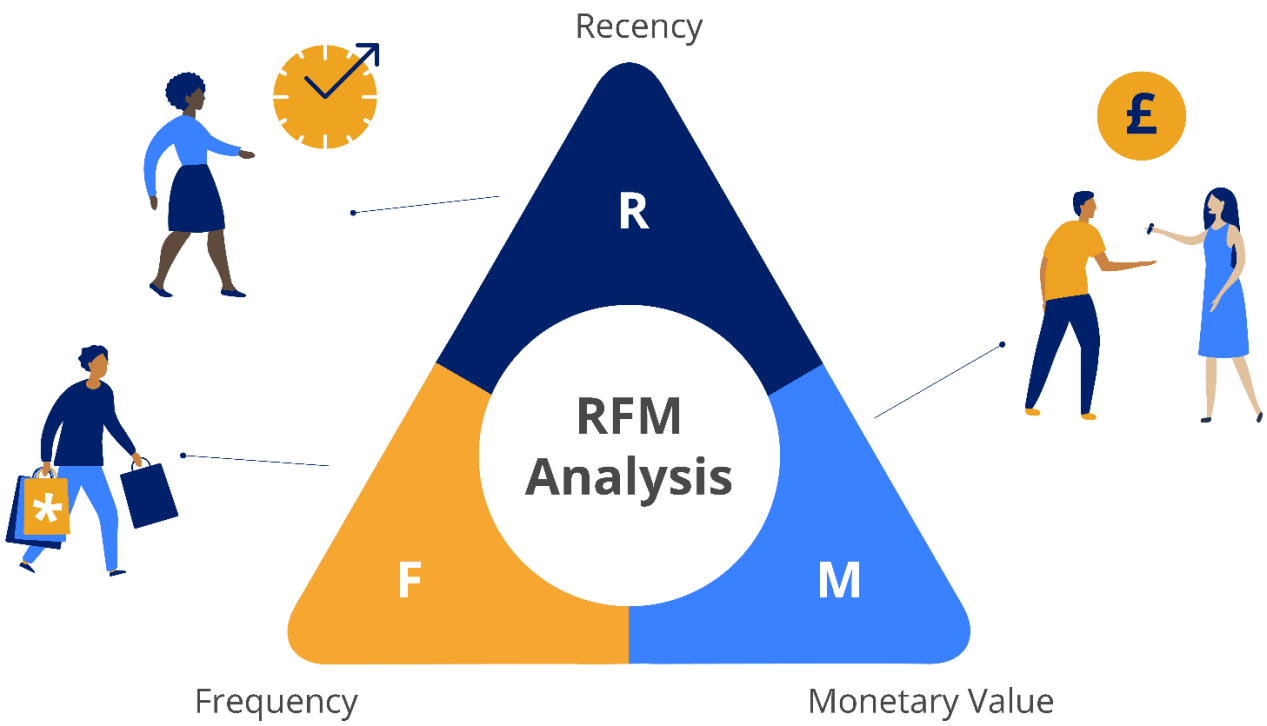
- **Higher Revenue Growth**: By understanding customers deeply, businesses can sell more and keep customers longer. Happy and loyal customers spend more over time. This leads to higher profits and steady business growth.

3. RFM Analysis

RFM analysis stands for Recency, Frequency and Monetary value. It helps businesses find their most important customers by looking at how they buy and spend. This analysis makes it easier to create personalized marketing plans that increase customer loyalty and lifetime value.

RFM shows which customers are worth investing in, which ones need more attention, and which ones are less important for business growth. Each part tells something different about customer behavior:

- **Recency**: How recently a customer bought something. Customers who bought recently are more interested and more likely to respond to new offers or promotions.
- **Frequency**: How often a customer buys. Frequent buyers are more loyal and engaged. They can be rewarded with loyalty programs or special discounts.
- **Monetary**: How much money a customer spends. Big spenders are very valuable for the business. They help to drive revenue and can be given special rewards or privileges.



4. Benefits of RFM Analysis in Customer 360

- **Better Customer Segmentation**: RFM helps group customers based on how recently, how often, and how much they buy. This makes it easier to understand different types of customers and treat them accordingly.

- **Personalized Marketing**: With RFM insights, businesses can send the right messages to the right people. For example, loyal customers can get special offers, while inactive ones can receive win-back campaigns.

- **Improved Customer Retention**: RFM helps find customers who may stop buying soon. Companies can take action early—such as giving discounts or reminders— to keep them engaged.

- **Higher ROI on Marketing**: By focusing marketing efforts on the most valuable customers, businesses save time and money. This leads to more effective campaigns and better results.

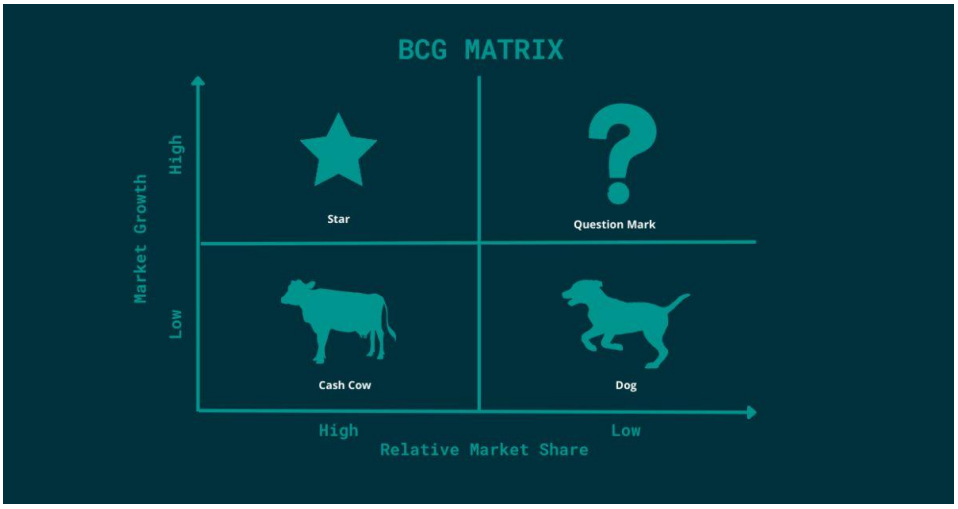
- **Data-Driven Decisions**: RFM provides clear, measurable insights about customer behavior. Teams can make smarter decisions about promotions, product focus, and customer service priorities.

- **Supports the Customer 360 View**: RFM adds valuable behavioral data to the Customer 360 system. It helps build a full picture of each customer's value, loyalty, and potential for future growth.

5. BGC Matrix in Customer 360

BCG Matrix (also known as the Boston Consulting Group analysis, the Growth-Share matrix, the Boston Box or Product Portfolio matrix) is a tool used in corporate strategy to analyse business units or product lines. The main purpose of the BCG Matrix is therefore to make investment decisions on a corporate level. Depending on how well the unit and the industry is doing, four different category labels can be attributed to each unit:

- **Star (VIP)**: They buy often, spend a lot, and are highly engaged with the brand. Businesses should focus on keeping them loyal through rewards, exclusive offers, or personalized experiences. If treated well, they can become long-term brand advocates.
- **Cash Cow (Loyal Customer)**: They already trust the brand and continue to purchase regularly, though their growth may have slowed. The main goal is to maintain satisfaction and loyalty through consistent quality and service. These customers help fund other growth opportunities in the business.
- **Question Mark (Potential Customer)**: They might be new customers or occasional buyers who are still learning about the brand. With the right marketing, discounts, or engagement efforts, they can become loyal and high- value customers. Businesses should test different strategies to discover what keeps them coming back.
- **Dog (Secondary Customer):** They rarely buy and may not be very interested in the brand anymore. Companies should avoid spending too many resources on them and focus on more promising segments instead. Sometimes, it's best to keep basic communication but shift attention to customers with higher value.



II. DETERMINE CUSTOMER SEGMENT

1. Introducing database to RFM Analysis

In this project, we have two tables to analyze:

Customer_Transaction Table

This table shows transaction of customers from June 2022 to August 2022 and has 4 columns:

Column Name	#	Data Type	Not Null	Auto Increment	Key	Default
123 Transaction_	1	int	[ ]	[ ]		
123 CustomerID	2	int	[ ]	[ ]		
A-Z Purchase_Da	3	varchar(50)	[ ]	[ ]		
123 GMV	4	int	[ ]	[ ]		

Customer_Register Table

This table shows information of customers who have registered for membership cards and has 7 columns:

Column Name	#	Data Type	Not Null	Auto Increment	Key	Default
123 ID	1	int	[ ]	[ ]		
A-Z Contract	2	varchar(50)	[ ]	[ ]		
123 LocationID	3	int	[ ]	[ ]		
123 BranchCode	4	int	[ ]	[ ]		
123 Status	5	int	[ ]	[ ]		
A-Z created_date	6	varchar(50)	[ ]	[ ]		
A-Z stopdate	7	varchar(50)	[ ]	[ ]		

2. Describe the steps to build RFM and customer segmentation

Tool: MySQL
how to do:

- Process OTLP data to OLAP from Customer_Transaction table
- Join Customer_Transaction table with Customer_Register to get the necessary information
- Use functions: Windows funtions, Case when, Datediff funtion to calculate and define customer segment by RFM rule

Step 1: Calculate Recency, Frequency, Monetery values:

- **Receficy:** period from last purchase date to 1/9/2022 (reporting date)
- **Frequeficy:** Total purchases from June 2022 to August 2022. Then divide by the contract age of each customer.
- **Monetery:** Total revenue of each customer from June 2022 to August 2022. Then divide by the contract age of each customer.

Note: The contract age is calculated by the distance between the date of membership registration and the current time of September 1, 2022. The reason for calculating the contract age is to divide the average of each customer based on the same time reference system (in this case, we calculate the contract age in years)

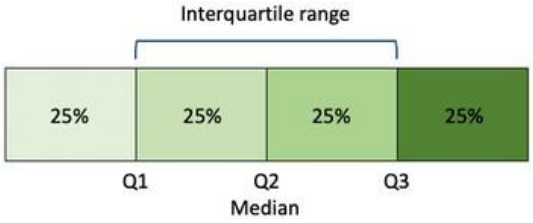
Calculate Recency, Frequency, Monetery values by CustomerID

	123 CustomerID	123 recency	123 frequency	123 monetary
1	1,039,378	92	0.28	0
2	982,647	92	0.27	0
3	995,756	92	0.27	0
4	1,028,296	92	0.27	0
5	471,369	92	0.19	0
6	516,857	92	0.2	0
7	520,757	92	0.2	0
8	520,957	92	0.2	0
9	529,009	92	0.2	0
10	533,931	92	0.21	0
11	534,931	92	0.21	0
12	538,740	92	0.21	0
13	549,751	92	0.21	0
14	125,201	92	0.14	0
15	156,041	92	0.15	0
16	180,263	92	0.15	0
17	195,143	92	0.15	0
18	206,944	92	0.15	0
19	209,608	92	0.15	0
20	209,658	92	0.15	0
21	214,041	92	0.15	0
22	214,446	92	0.15	0
23	214,486	92	0.15	0
24	232,725	92	0.16	0
25	232,735	92	0.16	0
26	243,749	92	0.16	0
27	250,050	92	0.16	0

3. Describe the steps to build RFM and customer segmentation

Step 2: Grouping based on RFM score using IQR method:

- Ranking for each index R,F and M
- Then classify 4 groups for each index using IQR Method (1 : worst - 4 best)



123 CustomerID	123 recency	123 frequency	123 monetary	123 R	123 F	123 M	A-Z RFM
1,039,378	92	0.28	0	1	4	1	141
982,647	92	0.27	0	1	4	1	141
995,756	92	0.27	0	1	4	1	141
1,028,296	92	0.27	0	1	4	1	141
471,369	92	0.19	0	1	2	1	121
516,857	92	0.2	0	1	2	1	121
520,757	92	0.2	0	1	2	1	121
520,957	92	0.2	0	1	2	1	121
529,009	92	0.2	0	1	2	1	121
533,931	92	0.21	0	1	2	1	121
534,931	92	0.21	0	1	2	1	121
538,740	92	0.21	0	1	2	1	121
549,751	92	0.21	0	1	2	1	121
125,201	92	0.14	0	1	1	1	111
156,041	92	0.15	0	1	1	1	111
180,263	92	0.15	0	1	1	1	111
195,143	92	0.15	0	1	1	1	111
206,944	92	0.15	0	1	1	1	111
209,608	92	0.15	0	1	1	1	111
209,658	92	0.15	0	1	1	1	111
214,041	92	0.15	0	1	1	1	111
214,446	92	0.15	0	1	1	1	111

Calculate RFM Score by CustomerID

4. Describe the steps to build RFM and customer segmentation

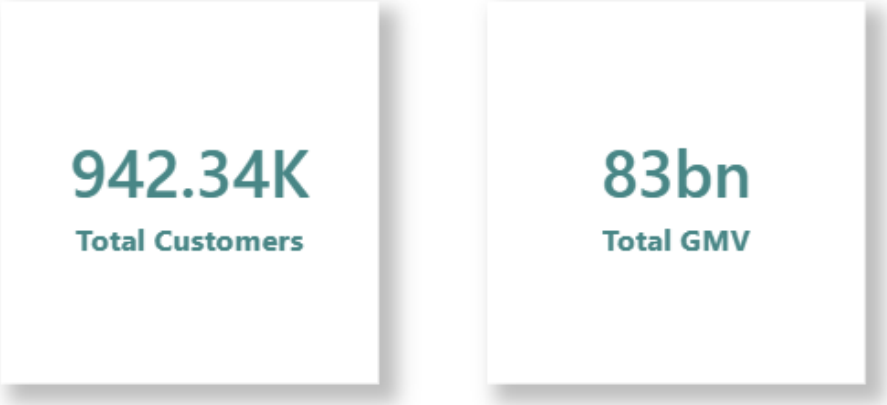
Step 3: Grouping based on RFM score using IQR method:

Customer Segment	RFM Score	Description
VIP	444, 443, 434, 433, 344, 343	Recent, frequent, high spenders – your best, most loyal customers.
Loyal	442, 432, 333, 323, 244, 243, 234, 233, 224	Buy often and recently, spend moderately high – regular and reliable customers.
Potential	441, 431, 424, 423, 342, 341, 332, 331, 324, 322, 242, 232	New or returning customers with good potential to become loyal
Recent Customers	422, 421, 414, 413, 412, 411	Recently started buying, low frequency and spend – new relationships to build on.
At Risk	321, 314, 313, 312, 311, 231, 223, 222	Haven't purchased recently, frequency or spending is dropping – re engagement needed.
Hibernating	241, 221, 142, 141, 132, 131, 124, 123, 122	Long inactive, low spend and frequency – nearly lost customers.
Lost	214, 213, 212, 211, 121, 114, 113, 112, 111	Lowest recency, frequency and monetary.

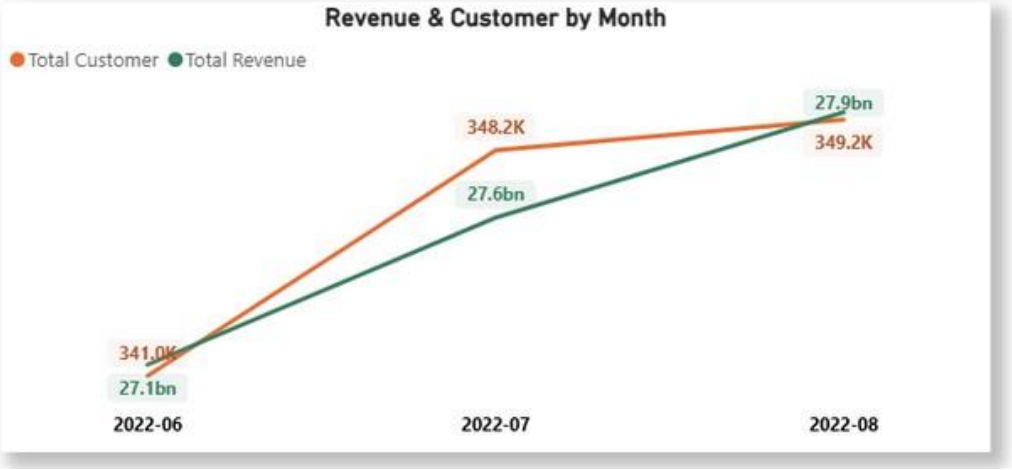
III. CUSTOMER ANALYSIS

1. Overall of customer

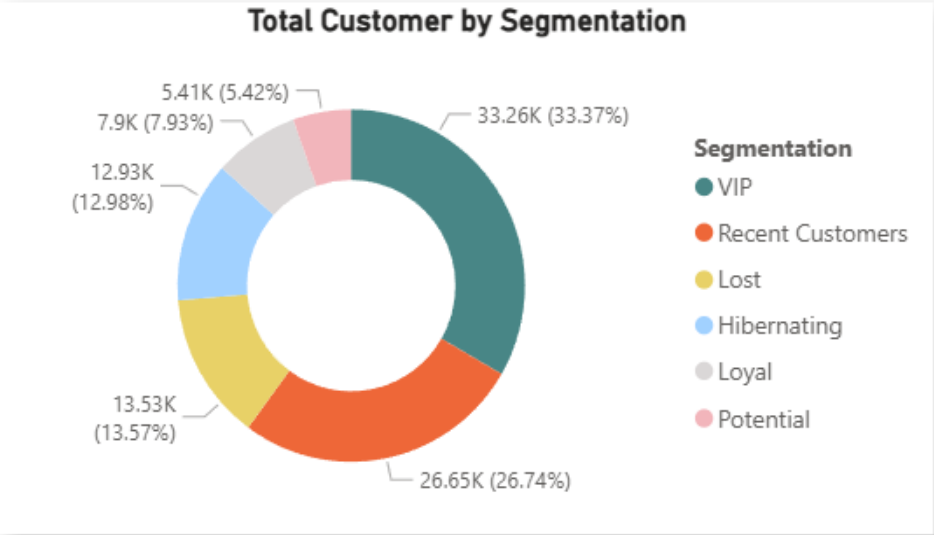
From June 2022 to August 2022, we served over 942K customers and generated a total revenue of 83 billion.



From June to August 2022, the number of active customers increased by 2.4% (from 341K to 349.2K), leading to a 3% rise in total revenue (from 27.1B to 27.9B). To drive stronger growth in the year-end period, a deeper customer segmentation analysis and targeted strategies are needed.



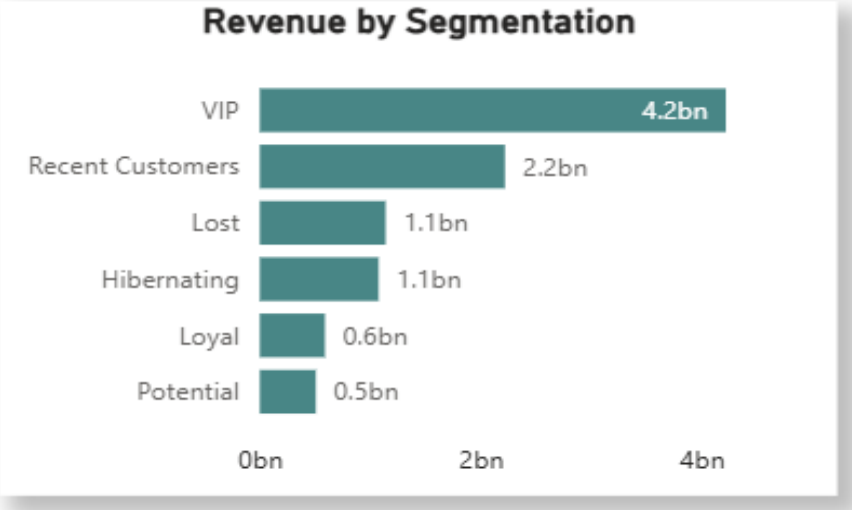
2. Customer segment analysis:



The VIP customer segment accounts for the highest proportion (approximately 33.37%), indicating that the company's products/services have strong retention power. However, contrary to expectations, the Potential and Loyal customer segments represent the lowest proportions (5.42% and 7.93%, respectively). This suggests that the company may not be effectively executing its sales and customer development strategies.

Further supporting this observation, the Lost customer segment—those who have not used the company's products or services for an extended period—ranks third in proportion at 13.57%. Identifying the reasons behind this churn and improving customer retention efforts should be a priority.

Meanwhile, the Recent Customers segment accounts for the second-largest share (26.74%), representing a significant opportunity. With appropriate promotional campaigns and attractive incentives, this group can be nurtured into Potential and VIP customers in the future. Effectively developing this segment will play a crucial role in shaping the company's long-term growth



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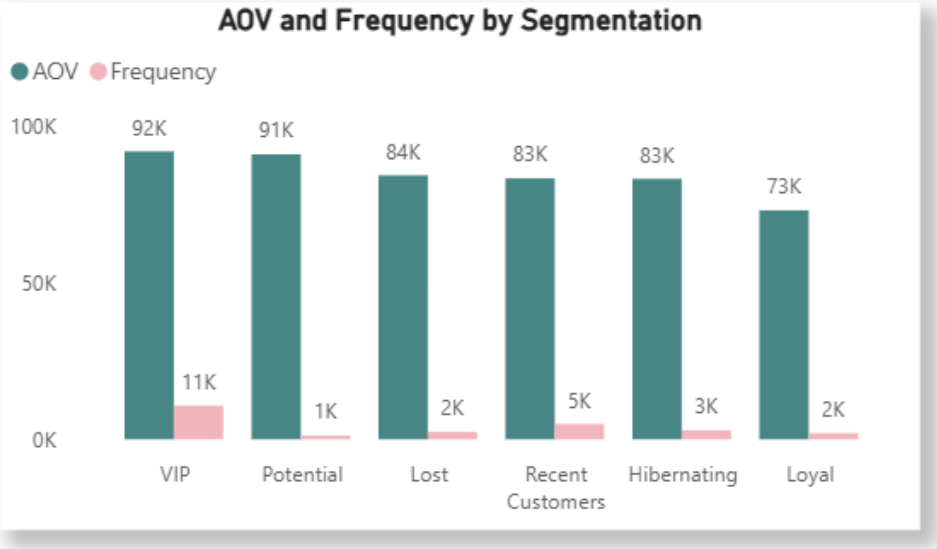

Revenue tells a very different story from AOV. VIP dominates at 4.2bn (43% of total), while Lost + Hibernating combined equal 2.2bn — the same as Recent Customers — despite being inactive. Loyal and Potential contribute the least (0.6bn and 0.5bn), each for opposite reasons: Loyal has volume but low AOV; Potential has high AOV but low volume.

AOV across all segments is relatively consistent (73K–92K), with VIP highest at 92K and Loyal lowest at 73K. The standout anomaly is Potential, whose AOV nearly matches VIP (91K) but has the lowest frequency (1K) — signaling high spending willingness but low retention. Meanwhile, Lost and Hibernating carry higher AOV than Loyal (84K, 83K vs 73K), suggesting they left for reasons other than budget.

=> While AOV remains stable across segments, it is frequency and customer volume that ultimately determine revenue outcome. To sustain growth, the business must focus on three things: protecting the VIP base, reactivating Lost and Hibernating customers who still carry strong spending power, and converting Potential into repeat buyers before that opportunity is lost.

Specifically, we will now explore the meaning of the four groups according to the BCG Matrix – the Growth – Share Matrix

Star (VIP Customers)

The Star group in the BCG Matrix here refers to the VIP Customers. They have a high purchase frequency, a very recent purchase, and are particularly generous in spending money with the company. Currently, this customer group comprises over 33.26K people (33.37%) and ranks at the top compared to other groups.

Question Mark (Potential Customers)

The Question Mark group includes Recent and Potential customers, who show initial engagement but have not yet developed strong loyalty or spending behavior. This segment accounts for 32.16% of the total customer base, including Recent Customers (26.74%) and Potential Customers (5.42%).

This represents a key growth opportunity, as effective marketing and engagement strategies can convert these customers into Loyal and VIP segments in the future.

Cash Cow (General Customers)

The Cash Cow group refers to Loyal customers. This segment demonstrates relatively stable purchasing behavior and contributes consistent revenue to the business. However, the proportion of this group remains relatively low at 7.93% (7.9K customers). Therefore, the company should focus on maintaining satisfaction and increasing engagement to further maximize revenue from this segment and prevent them from slipping into lower-value groups.

Dog (Secondary Customers)

The Question Mark group in the BCG Matrix here includes customer groups such as: Lost, Hibernating. These are two customer groups with low purchase frequency, almost no recent purchases, and extremely low revenue from customers. However, they currently represent a relatively significant proportion of the company's customer base. Specifically:

Lost: 13.57% (13.53K customers)
Hibernating: 12.98% (12.93K customers)

IV. SOLUTIONS

- **Star — VIP Customers:** Keep them engaged with exclusive rewards, early access, and personalized service. Strengthen emotional connection to turn loyalty into advocacy.
- **Cash Cow — Loyal Customers:** Maintain satisfaction through consistent quality, simple loyalty programs, and appreciation messages. Focus on long-term retention rather than discounts.
- **Question Mark — Potential Customers:** Use targeted promotions, onboarding emails, and personalized recommendations to drive repeat purchases. Test campaigns to identify what motivates conversion.
- **At Risk Customers:** Re-engage with win-back offers, reminder messages, or time-limited deals. Use behavioral data to personalize outreach and rebuild interest.
- **Hibernating Customers:** Reconnect through low-cost channels like email or social media (digital marketing). Offer small incentives or product updates to spark curiosity and bring them back.
- **Lost Customers**: Reach out through personalized SMS or phone calls to share exclusive offers or product updates. Create a sense of attention and care to encourage them to return. Keep communication occasional but genuine.

SQL CODE

```sql
with customer_satistics as (
    select
        ct.CustomerID,
        datediff("2022-09-01", max(str_to_date(ct.Purchase_Date, '%m/%d/%Y'))) as recency,
        round( count(distinct ct.Transaction) / (TIMESTAMPDIFF(month, str_to_date(cr.created_date, '%m/%d/%Y'), "2022-09-01") / 12),2) as frequency,
        round(sum(ct.GMV) / (TIMESTAMPDIFF(month, str_to_date(cr.created_date, '%m/%d/%Y'), "2022-09-01") / 12), 2) as monetary
    from customer_transaction ct
    join customer_registered cr on cr.ID = ct.CustomerID
    where ct.Purchase_Date  is not null
    group by ct.CustomerID
),

number_RFM as (
    select
        cs.*,
        row_number() over (order by cs.recency desc) as rank_R,
        row_number() over (order by cs.frequency asc) as rank_F,
        row_number() over (order by cs.monetary asc) as rank_M
    from customer_satistics cs
),
```

```sql
customer_RFM as (
    select
        CustomerID, recency, frequency, monetary,
        case
            when min(n.recency) <= n.recency and n.recency < (
                select cs.recency from number_RFM cs where cs.rank_R = (select round(max(rank_R)*0.25,0) from number_RFM)
            ) then 4
            when (
                select cs.recency from number_RFM cs where cs.rank_R = (select round(max(rank_R)*0.25,0) from number_RFM)
            ) <= n.recency and n.recency < (
                select cs.recency from number_RFM cs where cs.rank_R = (select round(max(rank_R)*0.5,0) from number_RFM)
            ) then 3
            when (
                select cs.recency from number_RFM cs where cs.rank_R = (select round(max(rank_R)*0.5,0) from number_RFM)
            ) <= n.recency and n.recency < (
                select cs.recency from number_RFM cs where cs.rank_R = (select round(max(rank_R)*0.75,0) from number_RFM)
            ) then 2
            else 1
        end as "R",
        case
            when min(n.frequency) <= n.frequency and n.frequency < (
                select cs.frequency from number_RFM cs where cs.rank_F = (select round(max(rank_F)*0.25,0) from number_RFM)
            ) then 1
            when (
                select cs.frequency from number_RFM cs where cs.rank_F = (select round(max(rank_F)*0.25,0) from number_RFM)
            ) <= n.frequency and n.frequency < (
                select cs.frequency from number_RFM cs where cs.rank_F = (select round(max(rank_F)*0.5,0) from number_RFM)
            ) then 2
            when (
                select cs.frequency from number_RFM cs where cs.rank_F = (select round(max(rank_F)*0.5,0) from number_RFM)
            ) <= n.frequency and n.frequency < (
                select cs.frequency from number_RFM cs where cs.rank_F = (select round(max(rank_F)*0.75,0) from number_RFM)
            ) then 3
            else 4
        end as "F",
        case
            when min(n.monetary) <= n.monetary and n.monetary < (
                select cs.monetary from number_RFM cs where cs.rank_M = (select round(max(rank_M)*0.25,0) from number_RFM)
            ) then 1
            when (
                select cs.monetary from number_RFM cs where cs.rank_M = (select round(max(rank_M)*0.25,0) from number_RFM)
            ) <= n.monetary and n.monetary < (
                select cs.monetary from number_RFM cs where cs.rank_M = (select round(max(rank_M)*0.5,0) from number_RFM)
            ) then 2
            when (
                select cs.monetary from number_RFM cs where cs.rank_M = (select round(max(rank_M)*0.5,0) from number_RFM)
            ) <= n.monetary and n.monetary < (
                select cs.monetary from number_RFM cs where cs.rank_M = (select round(max(rank_M)*0.75,0) from number_RFM)
            ) then 3
            else 4
        end as "M"
    from number_RFM as n
    group by CustomerID
    order by recency desc, monetary asc
)
select
    *, concat(R,F,M) as RFM
from customer_RFM
```